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Assignment #4

Dataset: BreastCancer from mlbench package

Objectives

This project uses different classification models to the BreastCancer dataset form mlbench package to predict tumor malignancy using patient features. The analysis uses machine learning workflow using performanceEstimation package.

Data Processing and Modeling Approach

The dataset was first cleaned by converting all factor variables except the target variable, Class, into numeric form. Missing values in the *Bare.nuclei* column were replaced with the mean. The *Id* column was removed because it does not contribute to prediction.

Five models used are:

1. Logistic Regression
2. Neural Network
3. Random Forest
4. Support Vector Machine (SVM)
5. Gradient Boosting (GBM)

The modeling task was defined using *PredTask*(), and each model's performance was estimated using 10-fold cross-validation repeated twice, evaluating four metrics - accuracy, precision, recall and F1 score.

Accuracy: The proportion of correct predictions over total predictions. Higher accuracy is good but can be misleading if the model is judged only based of accuracy.

Precision: Proportion of predicted positives that are actually positive. Higher precision is good, it means fewer false positives.

Recall: Proportion of actual positives correctly identified. High recall means fewer false negatives.

F1 Score: The mean of precision and recall. It balances both metrics and higher F1 is good.

Table 1 shows how well the models performed with four metrics it is evaluated with. It shows that Random Forest performed the best with an average F1 score of 0.974 followed very closely by SVM which achieved an F1 score of 0.974 as well. Although their

performance metrics are identical across accuracy, precision, recall and F1, Random Forest slightly performed better than SVM in terms of overall balance.

Gradient Boosting with 100 trees performed the third best with F1 score of 0.974, followed by Logistic Regression with F1 score of 0.973, and Gradient Boosting with 50 trees resulted F1 score of 0.970. Among the neural network variants, the model with 3 hidden units (size = 3) showed the best result with F1 score of 0.967 while the model with size = 2 showed the weakest performance due to high variance and precision.

Model <chr>	Accuracy <dbl>	Precision <dbl>	Recall <dbl>	F1_Score <dbl>
Random Forest	0.966	0.978	0.970	0.974
SVM	0.967	0.984	0.965	0.974
Gradient Boosting (100)	0.967	0.978	0.971	0.974
Logistic Regression	0.964	0.973	0.974	0.973
Gradient Boosting (50)	0.962	0.970	0.971	0.970
Neural Net (size=3)	0.957	0.970	0.965	0.967
Neural Net (size=1)	0.954	0.972	0.958	0.965
Neural Net (size=2)	0.841	0.852	0.978	0.899

8 rows

Table 1: Comparing model performance with four metrics

Figure 1, the boxplot visualizes the distribution of accuracy, recall, precision, and F1 score across 2x10 fold cross-validation for each classification model. Each red dot represents the performance on an individual fold, while black box shows the spread and central tendency of the scores.

We can see that SVM, Random Forest and GBM consistently achieved high median performance across all metrics, showing clusters of red dots near the top which indicates low variance and high stability. Logistic Regression and Neural Net also performed well but has more variation. Figure 1 shows that SVM, Random Forest, and GBM demonstrates strong consistency and robust generalization in cross-validation.

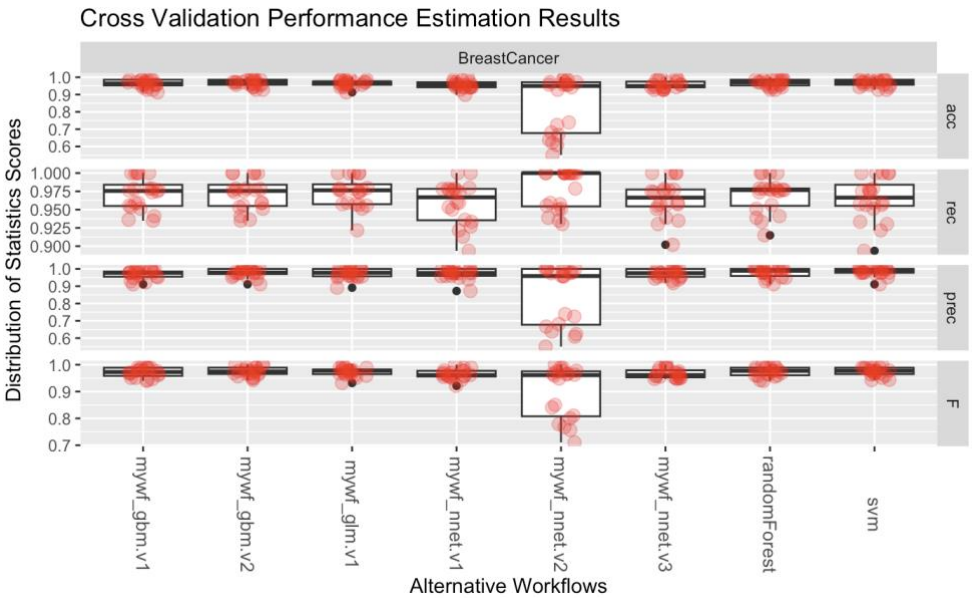

Figure 1: Cross Validation Performance Estimation Result

Since SVM, Random Forest, and Gradient Boosting had same F1 score, I created confusion matrix for all three models to see how they would perform in the final test set. To validate all models, the data set was retrained on a 70% training split and tested on the remaining 30%. Both SVM model and Random Forest achieved 100% accuracy on the test set, correctly classifying positives and negatives, however, Gradient Boosting showed a few misclassifications. Figure 2 shows the classification of training and testing data with 308 benign and 181 malignant in training data whereas 150 benign and 60 malignant in test data.

```
benign malignant
   308       181

benign malignant
   150        60
```

Figure 2: Classification of train and test data

Figure 3 and Figure 4 shows the confusion matrix for SVM and Random Forest model, correctly classifying the positives and negatives.

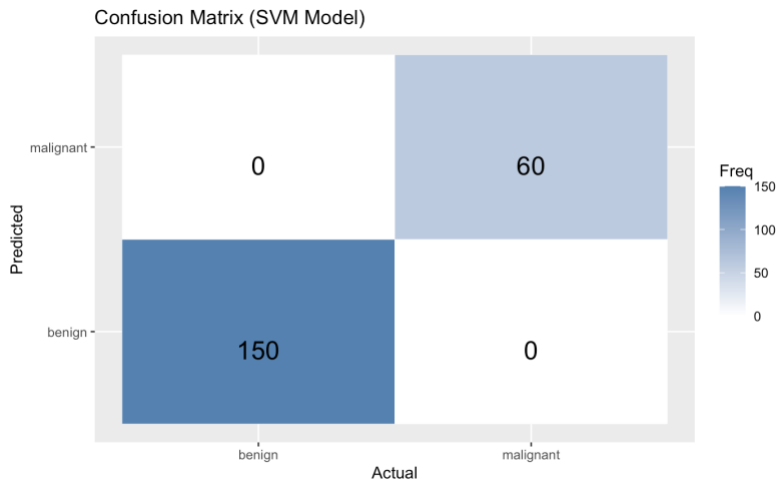

Figure 3: Confusion Matrix for SVM Model

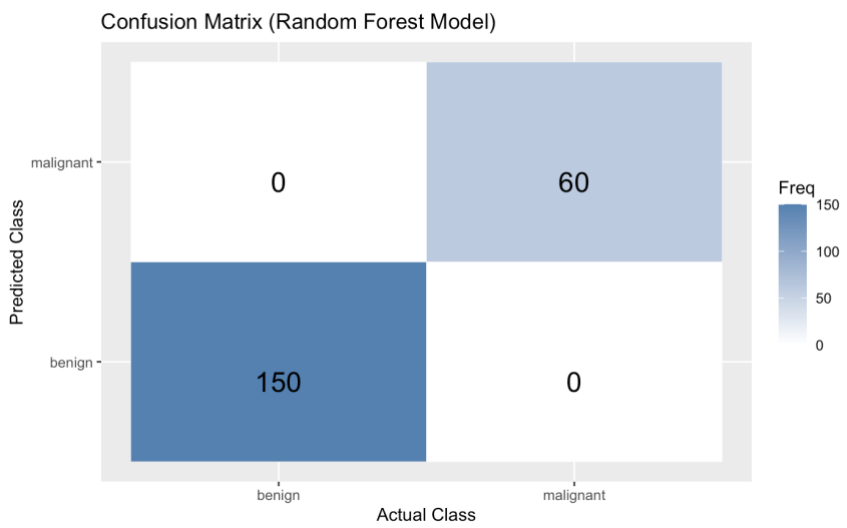

Figure 4: Confusion Matrix for Random Forest Model

Figure 5 shows GBM's confusion matrix which shows few misclassifications with 3 false negatives and 6 false positives.

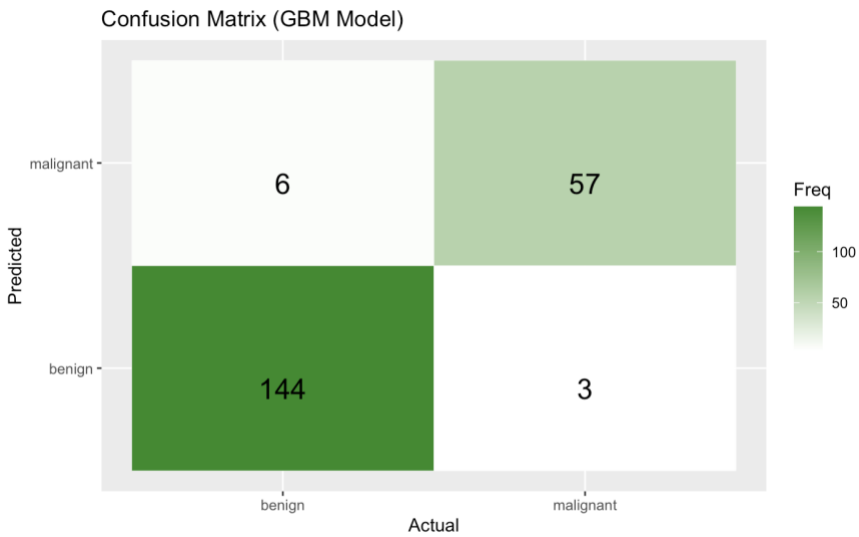

Figure 5: Confusion Matrix for Random Forest Model

Based on both cross-validation results and final test set performance, Support Vector Machine (SVM) and Random Forest are the top performing models. Each model achieved perfect classification on the test set, correctly identifying all benign and malignant cases with zero misclassifications with showing high accuracy, precision, recall, and F1 score during cross-validation. Gradient Boosting (GBM) also demonstrated excellent performance, especially the model with 100 trees, achieving F1 score nearly identical to SVM and Random Forest during cross-validation. However, GBM misclassified false negative and false positive in the second test set.

Since both SVM and Random Forest did really well, I wanted to compare the top predictors for both models.

	Importance <dbl>
Cell.shape	100.00000
Cell.size	99.36888
Bare.nuclei	88.55704
Bl.cromatin	86.98786
Epith.c.size	80.91806
Marg.adhesion	78.75569
Cl.thickness	74.69996
Normal.nucleoli	63.41564
Mitoses	0.00000

Figure 6: Top predictors for SVM

	MeanDecreaseGini
Cl.thickness	10.150922
Cell.size	76.361594
Cell.shape	38.469695
Marg.adhesion	7.012873
Epith.c.size	19.979170
Bare.nuclei	26.564861
Bl.cromatin	36.859056
Normal.nucleoli	11.684490
Mitoses	1.907061

Figure 7: Top predictors for Random Forest

While all three models are strong performers, Random Forest offers the best balance of cross-validation stability and test performance making it the best model for breast cancer classification on this dataset.